

POCKETSMART AI

Your Smart Budget & Recommendation Assistant

8-Phase Project Documentation

1. Brainstorming & Ideation Phase

Problem Identification

Users often find it difficult to manage budgets while selecting products and services for home decoration, parties, and personal styling. Multiple platforms and different price ranges make comparison time-consuming.

Proposed Idea

PocketSmart AI is a Generative AI-based budget and recommendation assistant that provides personalized suggestions according to the user's budget, preferences, occasion, and requirements.

Key Idea

The system combines budget planning, AI recommendations, product searches, and user preferences into a single platform.

Major Modules

- Home Interior Budget Planner
- Party & Event Budget Planner
- Jewelry & Outfit Recommendation Planner
- User Authentication
- Recommendation History
- AI Recommendation Engine

2. Requirement Analysis Phase

Functional Requirements

- User registration and login.
- Budget input and validation.
- Home interior planning.
- Party and event planning.
- Jewelry recommendation with optional outfit image.
- AI-generated recommendations.
- Recommendation history.
- Platform search links.
- User dashboard and personalized data.

Non-Functional Requirements

- Responsive and user-friendly interface.

- Secure authentication.
- Fast API response handling.
- Reliable budget calculations.
- Modular and maintainable architecture.
- Support for Windows laptop/local execution.

Technology Requirements

- Python
- FastAPI
- SQLite
- Jinja2
- HTML, CSS and JavaScript
- Google Gemini integration
- SQLAlchemy
- JWT authentication

3. Project Design Phase

System Architecture

User → Web Interface → FastAPI Backend → Recommendation Services → AI/Local Recommendation Engine → Database → Results

Main Components

Frontend: HTML/Jinja2 templates, CSS, JavaScript, planner forms, and recommendation display pages.

Backend: FastAPI, API routes, authentication, session handling, and request validation.

AI Layer: Gemini service, prompt generation, budget-aware recommendations, and optional image analysis.

Database: SQLite for user information and recommendation history.

Main API Modules

- /generate-home
- /generate-party
- /generate-jewelry
- /register
- /login
- /logout
- /history
- /session-info

4. Project Planning Phase

Development Plan

Phase	Major Activity
1	Problem identification and idea generation

2	Requirement collection and analysis
3	System and database design
4	Backend and frontend planning
5	Application development
6	Testing and optimization
7	Documentation preparation
8	Final demonstration

Module Priority

- Project setup
- Database and authentication
- Home Planner
- Party Planner
- Jewelry Planner
- AI recommendation service
- Recommendation history
- UI optimization and testing

Expected Outcome

A functional web-based budget recommendation system that can run locally on a Windows laptop.

5. Project Development Phase

Backend Development

The backend was developed using FastAPI with modular routing for different planner functions.

Frontend Development

- Home page
- Login
- Registration
- Dashboard
- Home Planner
- Party Planner
- Jewelry Planner
- Recommendation pages
- History

AI Development

The AI service processes user requirements and generates recommendations based on budget, category, preferences, and optional image information.

Database Development

SQLite is used to store user accounts and recommendation history.

Authentication

JWT-based authentication and password hashing are implemented for user security.

Platform Integration

The system generates search links for platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, CaratLane, and Myntra.

6. Project Testing Phase

Testing Areas

Functional Testing

- Registration and login
- Planner form submission
- Budget calculation
- Recommendation generation
- Image upload
- Recommendation history

Integration Testing

- Frontend-to-backend communication
- Backend-to-AI service communication
- Database operations
- Authentication and session handling

UI Testing

- Page navigation
- Form validation
- Responsive layout
- Recommendation display

Budget Testing

- Low-budget inputs
- Medium-budget inputs
- High-budget inputs
- Invalid and incomplete inputs

Expected Result

The system should generate relevant recommendations while maintaining the user's specified budget constraints.

7. Project Documentation Phase

Documentation Contents

1. Project Title
2. Problem Statement
3. Objectives

4. Proposed Solution
5. Requirements
6. Technology Stack
7. System Architecture
8. Module Description
9. Database Design
10. API/Route Description
11. Implementation Details
12. Testing Results
13. Screenshots
14. Advantages
15. Limitations
16. Future Enhancements
17. Conclusion

User Documentation

- Installing dependencies
- Configuring environment variables
- Starting the application
- Accessing the local website
- Using each planner
- Viewing recommendation history

8. Project Demonstration Phase

Demonstration Flow

Step 1 – Home Page: Introduce PocketSmart AI and its major features.

Step 2 – User Registration/Login: Demonstrate account creation and authentication.

Step 3 – Home Planner: Enter budget, room requirements, and preferences and generate recommendations.

Step 4 – Party Planner: Enter event type, guest count, venue requirements, and budget.

Step 5 – Jewelry Planner: Enter occasion, budget, and style preferences. Demonstrate optional outfit image analysis.

Step 6 – Recommendation Results: Show AI-generated recommendations and platform search links.

Step 7 – Dashboard & History: Demonstrate saved plans and previous recommendations.

Step 8 – Final Explanation: Explain how Generative AI, budget management, authentication, database storage, and recommendation services work together.

Final Outcome

PocketSmart AI provides a practical AI-assisted solution for budget-conscious planning across home interiors, events, and jewelry. It combines a FastAPI backend, AI recommendation services, database storage, authentication, and a user-friendly web interface.

Conclusion

PocketSmart AI is designed as a modular Generative AI application that simplifies budget planning and personalized recommendations. The eight-phase development process provides a structured path from idea generation and requirements analysis to development, testing, documentation, and final demonstration.